

Ozobot Prototype

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Overview & Purpose

This prototype documents the **Ozobot-based exploration** within the Wellbotics project. Unlike the Bee-Bot Garden Prototype, which represents a **fully integrated game system**, the Ozobot work functions as a **design exploration and feasibility study** focused on understanding how **robot sensing, physical navigation, and environmental cues** might support **emotional recognition and regulation** in young learners.

The Ozobot prototype does not aim to finalize a specific game. Instead, it investigates:

- What Ozobots can technically do
- How those capabilities could map to **emotional states**
- How **world-building and spatial design** might scaffold SEL learning
- Where constraints (technical, economic, pedagogical) emerge

This section plays a critical role in helping the team **narrow scope, identify differentiation challenges**, and **refine learning goals**.

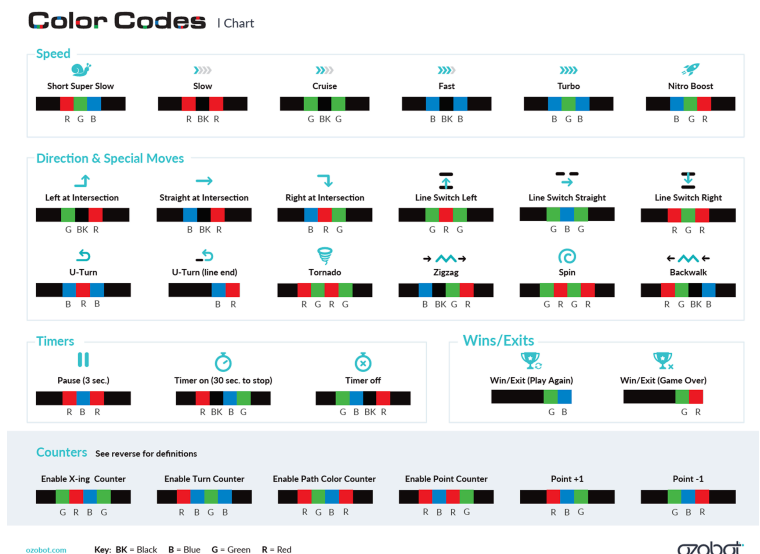
Technical Capabilities of Ozobot

Ozobot capabilities that informed ideation:

- Move
- Detect lines
- Take location points as instructions
- Change direction
- Control pace
- Emit sound
- Emit light
- Detect other Ozobots
- Detect color

These affordances position Ozobot as a robot that:

- Responds to **environmental input** rather than only explicit commands
- Can encode meaning through **color, speed, and proximity**
- Supports **multi-agent interaction**



This capability set made Ozobot an attractive platform for exploring **emotion-driven behavior**, where the robot's actions could *represent* emotional states rather than simply execute navigation commands.

Learning Focus & Conceptual Shift

From “Programming Movement” to “Programming Emotion”

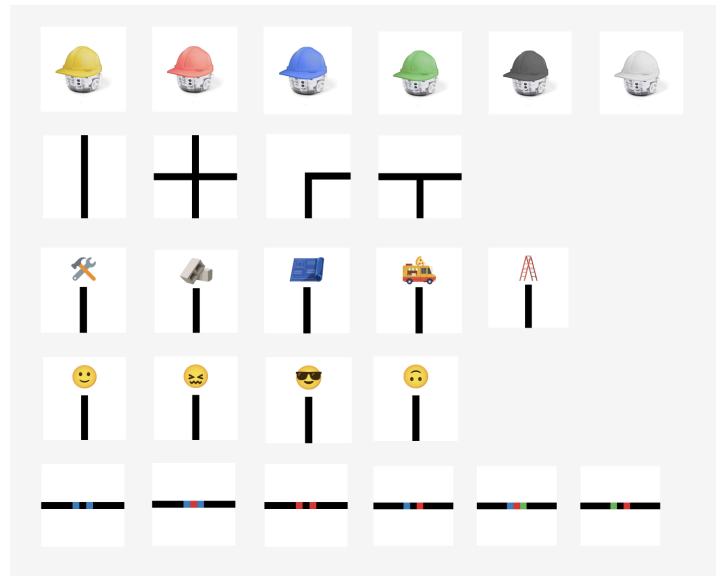
A key conceptual shift emerged during Ozobot exploration:

Rather than programming *where* the robot goes, the team explored programming

how the robot feels or behaves,

allowing:

- Speed to represent emotional intensity
- Color to represent emotional state
- Direction changes to reflect regulation or dysregulation
- Interaction with other robots to reflect social dynamics



This reframing moved the learning focus away from broad SEL and toward **emotional recognition and regulation**, which later became an explicit constraint agreed upon by the team.

Learning Domains Addressed

1. Emotional Recognition & Regulation (Primary Focus)

The Ozobot prototype emphasizes:

- Recognizing emotions (in self and others)
- Managing emotional responses
- Observing how emotional “states” change behavior

By externalizing emotion into robot behavior, learners can:

- Observe emotions as systems
- Discuss them with emotional distance
- Experiment safely with change and repair

This aligns directly with the team's later consensus to **tighten the learning goal** and avoid overly broad SEL framing.

2. Programming & Computational Thinking (Secondary)

While programming remains present, it is treated as a **means**, not an end:

- Sequencing
- Cause-and-effect reasoning
- Debugging emotional “misbehavior”
- Systems thinking (inputs → states → outputs)

Ozobot's line-following and color-detection capabilities support this without requiring abstract syntax.

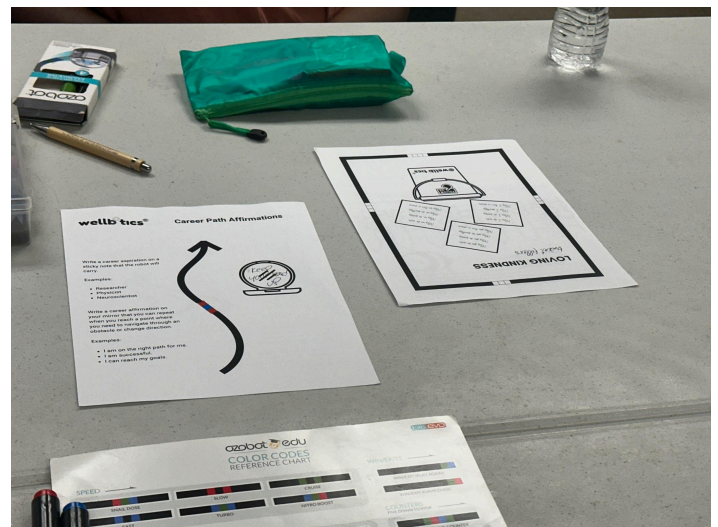
World-Building & Spatial Design Exploration

A major component of the Ozobot work involved **scenario and world design**, rather than final mechanics.

Explored World Concepts

From team brainstorming and visual mockups:

- **Pattern Language Framework** (emotional “centers,” boundaries, paths)
- **Calm Courtyard-style spaces** where emotional states are spatially encoded
- **Construction / Town-Building metaphors**
- **Floating islands** differentiated by color, speed, or rule sets
- **Carcassonne-style tile systems** where players collaboratively build the board

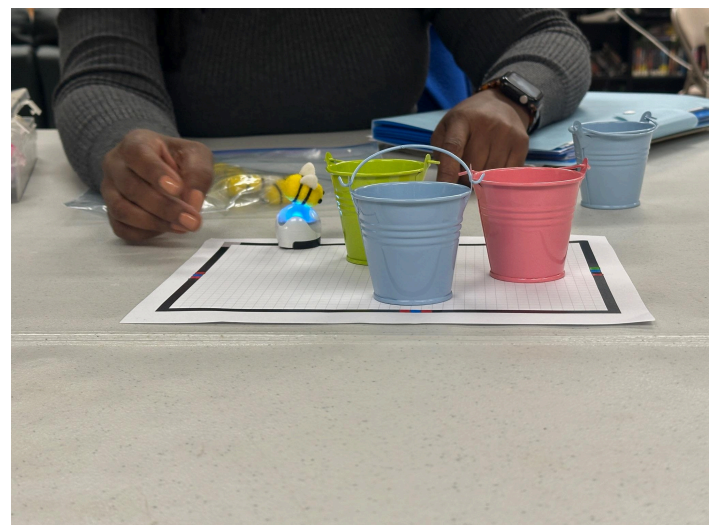


Across all concepts, the **“build the board” mechanic** emerged as a key throughline and later consensus constraint.

Session Structure & Gameplay Considerations

The Ozobot prototype explored:

- **Station-based learning** for younger learners (ages 5–6)
- **Multi-session structures** (2–3 sessions per experience)



- RPG-inspired progression (drawing from games like *Mice and Mystics* and *Gloomhaven*)
- Collaborative team roles rather than individual competition

These ideas remained exploratory and were not locked into a single rule system.

Key Discovery: Differentiation Constraint

A critical turning point occurred when the team discovered that **Ozobot already offers Challenge Mats** that closely resemble:

- Physical maps
- Mission-based navigation
- Coding challenges
- Tile-like layouts

This discovery prompted an explicit design reckoning:

- Ozobot's existing ecosystem overlaps heavily with the team's ideas
- Differentiation would need to come from **emotional framing, facilitation, or mechanics**, not surface form

This realization directly influenced later discussions about:

- Robot-agnostic design
- Custom hardware possibilities
- Shifting emphasis away from Ozobot as a core platform

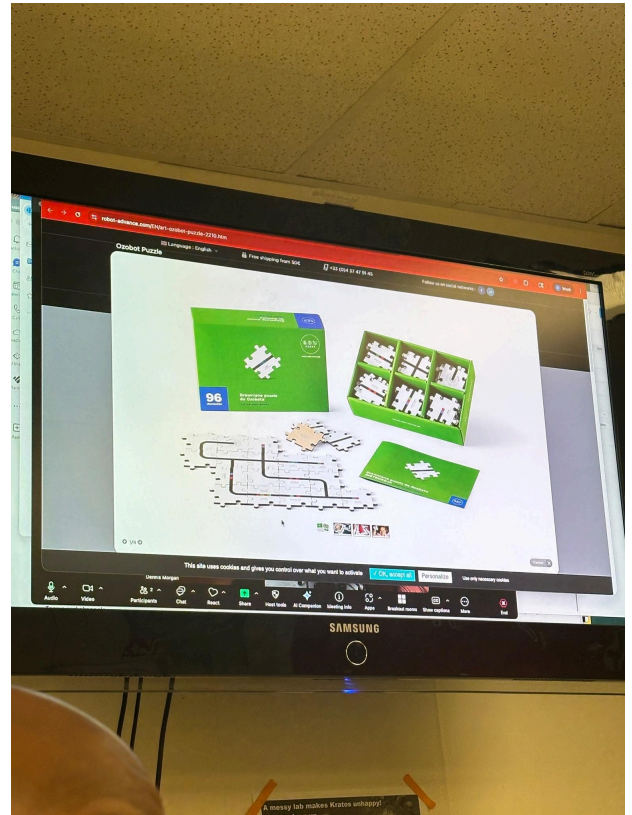
Constraints & Design Decisions Emerging

By the end of this phase, several **constraints and directions** were agreed upon:

- Narrow focus to **emotional recognition and regulation**
- Maintain **board-building** as a differentiator
- Avoid being locked to a single robot platform
- Consider ethical implications of AI integration for young children
- Reduce aesthetic exploration in favor of mechanics clarity

Status of the Ozobot Prototype

The Ozobot work is best understood as:



- A **mid-fidelity exploratory prototype**
- A **capability and constraint discovery phase**
- A **design decision catalyst**, not a final solution

It intentionally surfaces tensions between:

- Platform convenience vs. differentiation
- Emotional abstraction vs. physical embodiment
- Openness vs. curriculum clarity

Conclusion

The Ozobot prototype played a crucial role in the Wellbotics project by revealing both the **potential and limitations** of emotion-driven robotic learning. While Ozobot's sensing and feedback capabilities made it well-suited for exploring emotional representation through movement, color, and interaction, existing platform solutions constrained meaningful differentiation.

Rather than resulting in a finalized Ozobot game, this section clarified the project's **true design priorities**: emotional recognition and regulation, facilitated reflection, and collaborative world-building. These insights directly informed the shift toward a more flexible, facilitator-centered, and emotionally explicit system, culminating in the Bee-Bot Garden Prototype and the AI-supported exploration.

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